Timely and Effective Care Across U.S. States

Proposal

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```r
library(tidyverse)
```

Dataset

```r
care <- read_csv("data/care_state.csv")
state_pop <- read_csv("data/state_population.csv")
glimpse(care)
```

```
Rows: 1,232
Columns: 8
$ state        <chr> "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK~
$ condition    <chr> "Healthcare Personnel Vaccination", "Healthcare Personnel~
$ measure_id   <chr> "HCP_COVID_19", "IMM_3", "OP_18b", "OP_18b_HIGH_MIN", "OP~
$ measure_name <chr> "Percentage of healthcare personnel who are up to date wi~
$ score        <dbl> 7.3, 80.0, 140.0, 157.0, 136.0, 136.0, NA, 196.0, 230.0, ~
$ footnote     <chr> NA, NA, "25, 26", "25, 26", "25, 26", "25, 26", "25, 26",~
$ start_date   <date> 2024-01-01, 2023-10-01, 2023-04-01, 2023-04-01, 2023-04-~
$ end_date     <date> 2024-03-31, 2024-03-31, 2024-03-31, 2024-03-31, 2024-03-~
```

The dataset is the **Centers for Medicare & Medicaid Services (CMS) "Timely and Effective Care"** collection, sourced from data.cms.gov and curated for TidyTuesday week 14, 2025. It reports state-level hospital quality measures for reporting periods spanning 2023–2024. The file `care_state.csv` holds 1232 observations of 8 variables, covering 56 jurisdictions (50 states, DC, and U.S. territories) and 22 distinct measures grouped into 6 condition areas. The data are stored in a long/tidy layout, one row per state-and-measure, so each analysis pivots the relevant subset wider. The key variables are `state` and `condition` (categorical), `measure_id`/`measure_name` (the metric being reported), and `score` (the numerical value, whose units and direction differ by measure).

I chose this dataset because emergency-department crowding and hospital care quality are active national policy concerns, the data are recent and federally maintained, and the structure cleanly satisfies our need for both numerical (`score`, state population) and categorical (`condition`, Census `region`) variables. It also lets us recode a meaningful geographic grouping and merge in external population data to enrich the analysis.

I derive two variables used across the questions. `region` is the four-level U.S. Census region (Northeast, Midwest, South, West) recoded from each state code, and `state_pop` is the state's 2024 population, merged from a small Census-estimate file committed to `/data`.

Questions

Question 1 — Emergency-department throughput. How does the median time patients spend in the emergency department before leaving (`OP_18b`) vary by **ED volume tier** (low through very high, captured by the `OP_18b_*_MIN` measures) and **Census region**, and is it associated with **state population**?

Question 2 — Treatment effectiveness. Across **Census regions**, how do states compare on **treatment-effectiveness** measures, stroke CT timeliness (`OP_23`), colonoscopy follow-up appropriateness (`OP_29`), and safe opioid use (`SAFE_USE_OF_OPIOIDS`), and do states that score well on one tend to score well on the others?

These questions are distinct: Question 1 uses the ED-timing measures plus population, while Question 2 uses three effectiveness measures; they share only the derived `region` variable. Because the data are observational, we will frame all findings as **associations** and avoid causal language.

Analysis plan

Question 1 (ED throughput). Filter to `condition == "Emergency Department"`, keep the volume-stratified `OP_18b_LOW_MIN`, `OP_18b_MEDIUM_MIN`, `OP_18b_HIGH_MIN`, and `OP_18b_VERY_HIGH_MIN` measures, and pivot so each state has one wait-time value per volume tier. Join `region` and `state_pop`. Variables involved: ED wait time (numerical), volume tier (categorical), region (categorical), and state population (numerical), four variables. Planned `ggplot2` figures: (1) boxplots of wait time by volume tier, colored or faceted by region; (2) a scatter of state population versus median ED wait time.

Question 2 (treatment effectiveness). Filter to the three effectiveness measures, pivot them wide so each state has one column per measure, and join `region`. We will first confirm each measure's direction ("higher is better" versus "lower is better") from `measure_name`. Variables involved: `OP_23`, `OP_29`, and `SAFE_USE_OF_OPIOIDS` scores (numerical) plus region (categorical), more than two variables. Planned `ggplot2` figures: (1) a regional comparison of the measures (e.g., point-range or bar summaries by region); (2) a scatter relating two measures across states to assess whether strong performance on one accompanies strong performance on another.

Derived variables. `region` is built from base R's `state.abb/state.region` (recoding "North Central" to "Midwest" and assigning DC to the South). `state_pop` is merged from `data/state_population.csv`. Territories lacking a Census region or population estimate will be dropped from the relevant analyses and this exclusion will be reported.

Weekly plan of attack

Task Name	Status	Due	Priority	Summary
Finalize dataset & draft proposal	Complete	Jun 4	High	Load data, write Q1/Q2, complete proposal.qmd and data README
Submit proposal for peer review	Complete	Jun 4, 5pm	High	Render and push proposal
Address peer-review issues	Not started	Jun 9, 5pm	High	Revise proposal and close GitHub issues
Data wrangling (region, population, pivots)	Not started	Jun 13	High	Build clean tidy frames for both questions
Question 1 analysis & figures	Not started	Jun 18	Moderate	ED throughput plots and discussion
Question 2 analysis & figures	Not started	Jun 23	Moderate	Effectiveness plots and discussion
Write-up (index.qmd) & website	Not started	Jun 27	Moderate	Introduction, both questions, conclusions
Slides & recorded presentation	Not started	Jun 30, 5pm	High	~6 content slides, presentation under 5 minutes